

PLC-Based Automated Bottle Filling with AI-Based Quality Inspection

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Problem Statement & Major Problems

Develop an automated bottle filling and quality inspection system using PLC, YOLO, and OpenCV to reduce manual handling and inspection errors.

Major Problems

- Manual Handling – Low process efficiency
- Bottle Damage – Defective bottles may be missed
- Inconsistent Filling – Under/overfilling
- Water Spillage – Spillage may go unnoticed
- Human Error – Inconsistent inspection
- Lack of Automation – Limited vision-based inspection

OBJECTIVES

- Study the existing manual bottle-filling and inspection process and its limitations.
- Design and develop a PLC-based automated bottle-filling system.
- Develop a conveyor mechanism for automatic bottle movement and positioning.
- Implement bottle detection using a proximity sensor.
- Control the conveyor, water pump and filling sequence using PLC.
- Develop a camera-based bottle inspection system using YOLO and OpenCV.
- Detect damaged, cracked or deformed bottles using YOLO.
- Detect water level in the filled bottle using OpenCV.
- Detect visible water spillage using OpenCV-based image processing.
- Classify bottles based on inspection results as PASS/FAIL or GOOD/DEFECTIVE.
- Provide red-light indication when a defective bottle or filling-related problem is detected.
- Integrate the PLC, sensor, conveyor, pump, camera, AI and inspection system and validate the complete prototype.



SOFTWARE FOR AI & SYSTEM ARCHITECTURE

SOFTWARE + SYSTEM ARCHITECTURE

Software Required



Python 3.14.3
Programming language



Visual Studio Code
Development environment



OpenCV
Water-level and spillage inspection

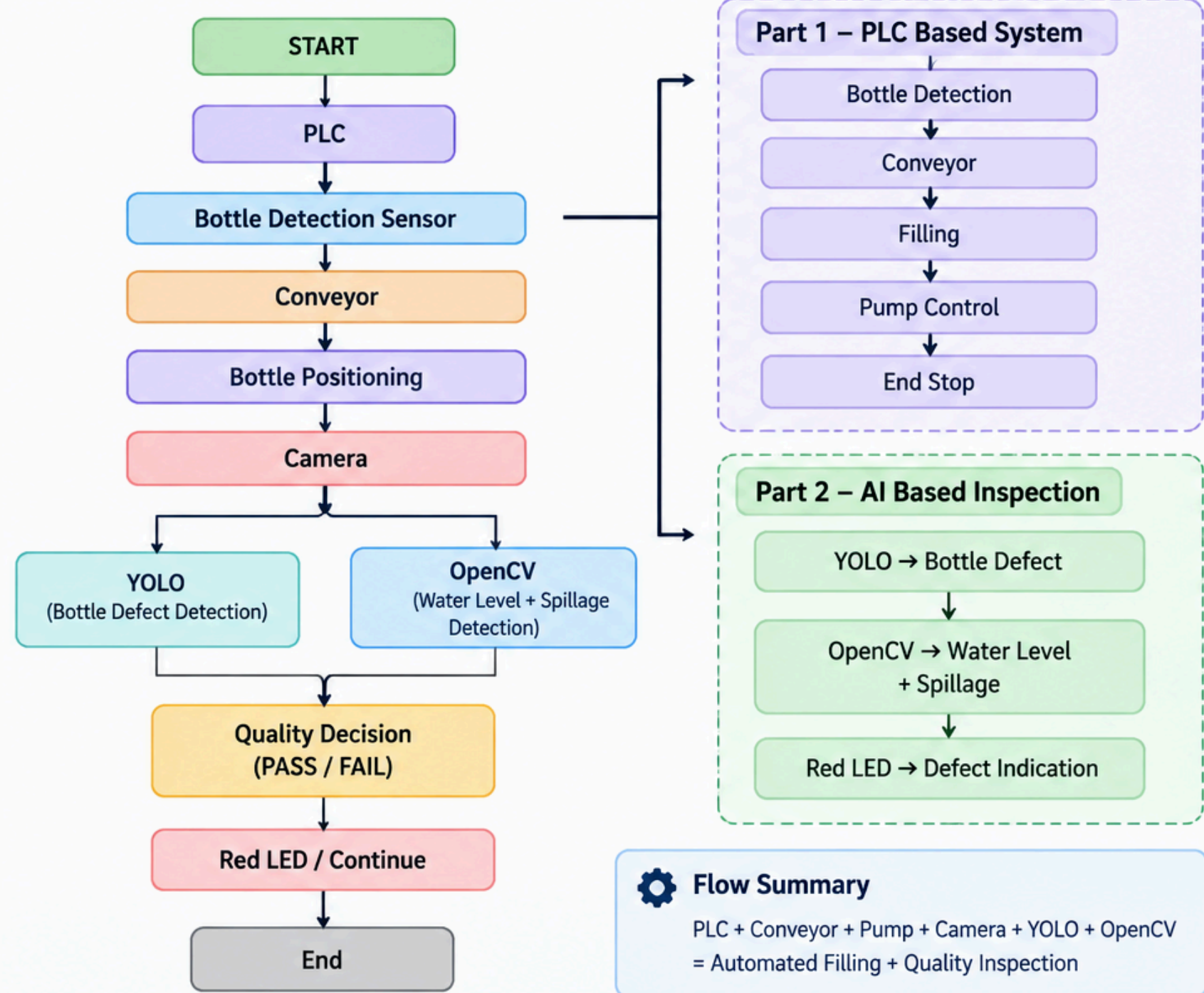


Ultralytics YOLO
Bottle defect detection



NumPy
Image/data processing

System Architecture



PLC I/O LIST – PART 1

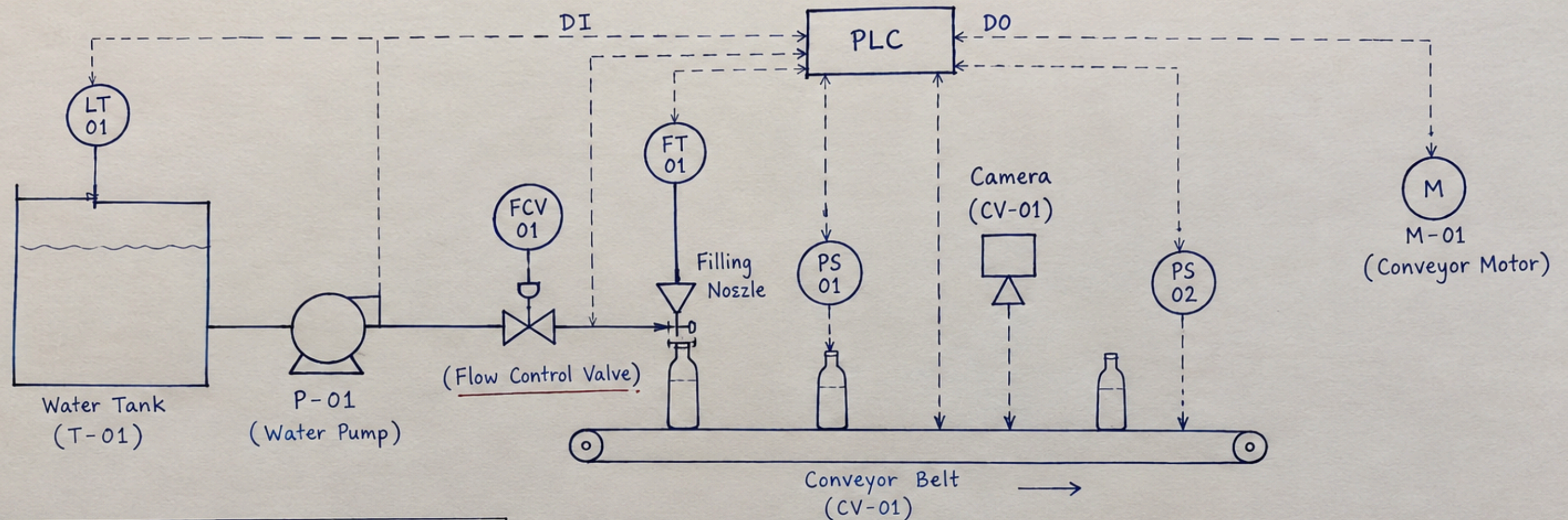
Tag	Description	Type	Signal	Address*
DI-01	Start Push Button	DI	NO	I0.0
DI-02	Stop Push Button	DI	NC	I0.1
DI-03	Bottle Detection Sensor	DI	NO/NC	I0.2
DO-01	Conveyor Motor	DO	—	Q0.0
DO-02	Water Pump	DO	—	Q0.1

** Addresses are provisional/symbolic and should be finalized according to the actual PLC model.*

AI INSPECTION LIST – PART 2

Tag	Description	Type	Function / Output
AI-01	USB Webcam	Input	Captures bottle image/video
AI-02	YOLO Model	AI	Good / Damaged / Deformed
AI-03	OpenCV – Water Level	Vision	Low / Correct / High
AI-04	OpenCV – Spillage Detection	Vision	Spillage / No Spillage
AI-05	Python Program	Processing	PASS / FAIL Decision
AI-06	USB-Controlled Relay	Output	ON / OFF control
AI-07	Red LED Indicator	Output	ON for defective/problem bottle

P&ID DIAGRAM



Instrument List / Tag List		
Tag No.	Description	Type
LT-01	Level Transmitter (Tank)	LT
FT-01	Flow Transmitter (Water)	FT
FCV-01	Flow Control Valve	FCV
PS-01	Bottle Detection Sensor (Start)	PS
PS-02	End Position Sensor	PS
CV-01	Camera	CV
M-01	Conveyor Motor	M
P-01	Water Pump	P

Legend	
	: Transmitter
	: Sensor
	: Motor
	: Pump
	: Control Valve
	: PLC
	: Signal Line
	: Process Line

Notes :

1. LT-01 measures water level in tank.
2. FT-01 measures flow rate of water.
3. FCV-01 is a flow control valve to maintain constant flow rate to the filling nozzle.
4. PS-01 detects bottle at filling position (start).
5. PS-02 detects end position of bottle.
6. CV-01 is used for AI inspection (Part 2).
7. M-01 drives the conveyor belt.
8. P-01 is the water pump.

* Flow control valve (FCV-01) is used to regulate the flow rate of water from the pump to the filling nozzle, so that each bottle gets the correct amount of water (filling time/quantity). Without it, the flow may be too high or too low, which can affect the filling level.



PROGRESS LEVEL 1



PROGRESS LEVEL 2

